Task 4: Session 4 (AIML) - Assignment Questions

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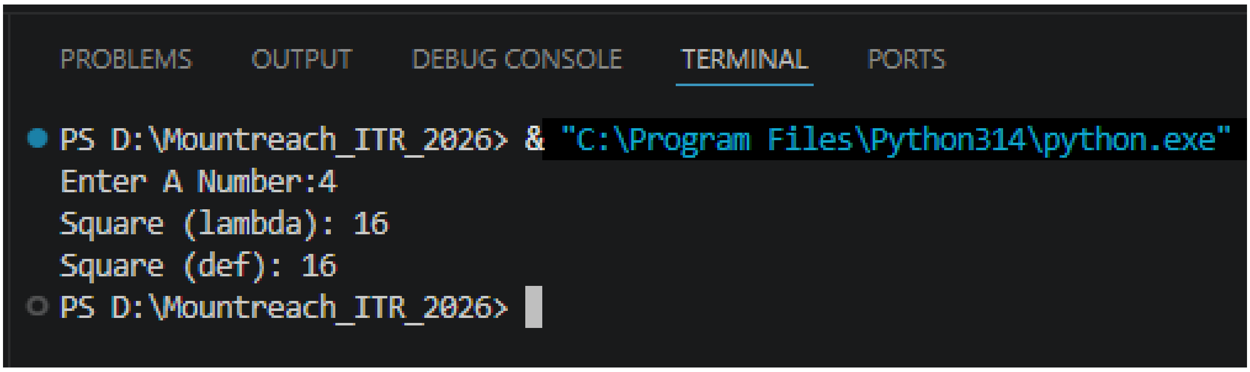
1. **Lambda Function**

   Write a lambda function named square that takes a number x and returns its square.

   Call the lambda function with input from the user and print the result.

   Also write the same logic using a normal def function and compare both in comments.

   Output:

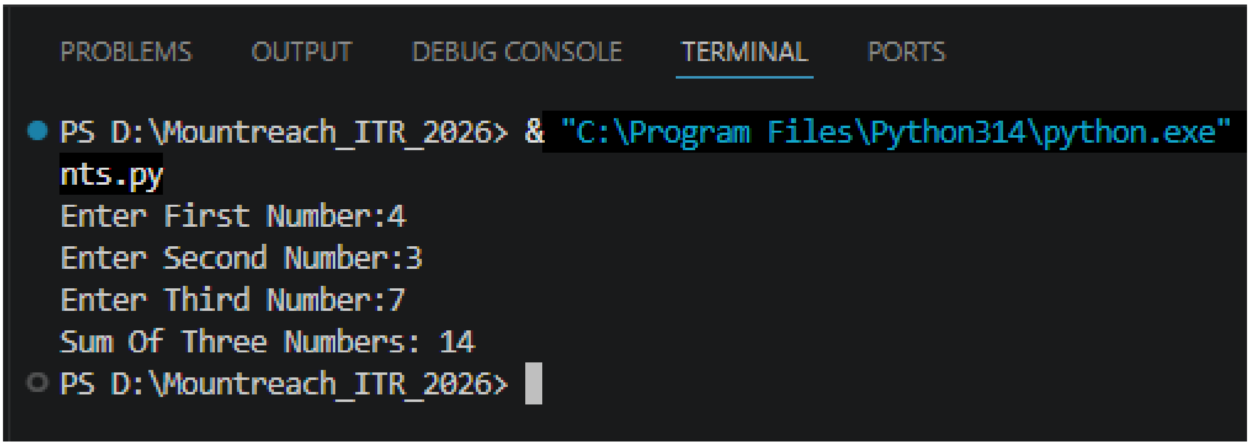


2. Lambda With Multiple Arguments

Create a lambda function add_three(a, b, c) that returns the sum of three numbers.

Take three numbers as input from the user and print their sum using this lambda function.

Output:



3. For Loop Revision

Write a program using a for loop and range() to:

- Print all numbers from 1 to 25.
- Print the sum of all numbers from 1 to 20.
- Print the table of 5 5 x 1 to 5 x 10.

Add comments explaining when to use for loop.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Numbers From 1 To 25:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Sum Of 1 To 20: 210

Multiplication Table Of 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
PS D:\Mountreach_ITR_2026>
```

4. While Loop Revision

Write a program using a while loop that:

- Repeatedly asks the user to enter a positive number.
- Keeps adding the numbers to a running total.
- Stops when the user enters 0 or a negative number.
- Finally prints the total sum.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"

Enter A Positive Number:8
Enter A Positive Number:7
Enter A Positive Number:6
Enter A Positive Number:-5

You Entered 0 Or A Negative Number.
Total: 21.0
PS D:\Mountreach_ITR_2026>
```

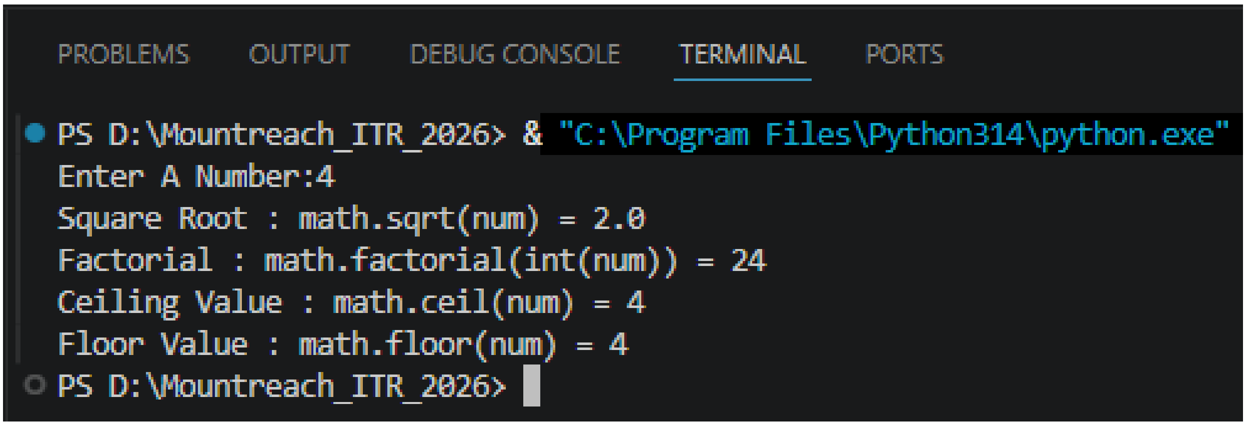
5. Math Module

Write a program that imports the math module and does the following:

- Takes a number as input.
- Prints its square root using math.sqrt().
- Prints its factorial using math.factorial() (for positive integers only).
- Prints the ceiling and floor value using math.ceil() and math.floor().

Add comments about the usefulness of the math module.

Output:



6. Random Module

Write a program using the random module that:

- Generates and prints 5 random integers between 1 and 100.
- Generates and prints one random number between 50 and 150.
- Prints a random floating point number between 0 and 1.

Use different functions from the random module.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Random Module Demo:
5 Random Integers (1-100):
70
18
11
63
67
Random Integer (50-150): 86
Random Float (0-1): 0.23605997604674449
PS D:\Mountreach_ITR_2026>
```

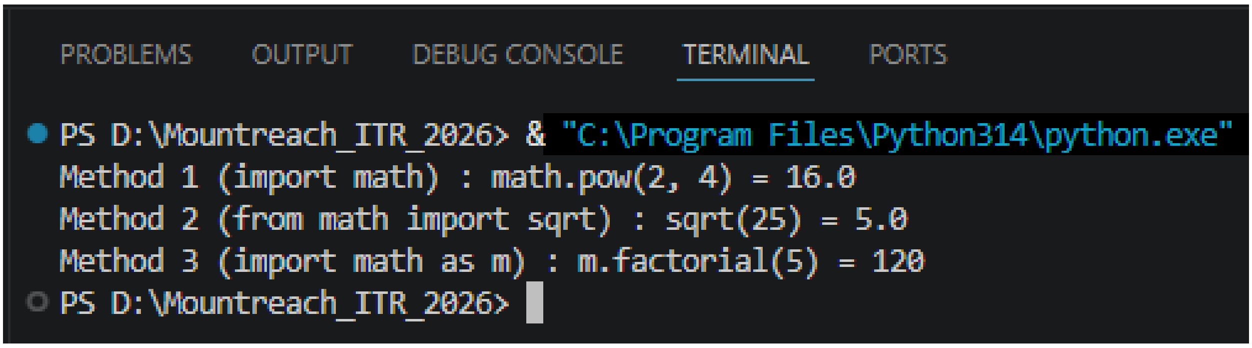
7. Import Methods

Demonstrate different ways of importing modules:

- Import the entire math module using import math and use math.pow(2, 4).
- Import only sqrt from math using from math import sqrt.
- Import math with an alias as m and use m.factorial(5).

Print the results with clear messages.

Output:



8. Custom Module

Create a custom module file named mymodule.py that contains:

- A function greet(name) that prints "Hello [name], Welcome to Python Class!"
- A function calculate_power(base, exp) that returns base raised to exponent.

Then write a main program that imports this module and uses both functions.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Custom Module Demo:
Hello Mihir Welcome to Python Class!
2 Raised To 8 = 256
PS D:\Mountreach_ITR_2026>
```

9. name == "main"

Write a program that:

- Creates a small function and demonstrates the use of if __name__ == "__main__": so that some code runs only when the file is executed directly.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
This File Is Being Run Directly.
Hello, Welcome To Python!
PS D:\Mountreach_ITR_2026>
```

10. Mini Project - Combined Concepts

Create a Simple Math Utility Program using functions, lambda, loops and modules:

Write the following:

- A lambda function to calculate square of a number.
- A normal function using math module to calculate power.
- Use a while loop to repeatedly ask the user for a number and choice Square / Power / Random Number).
- Use random module to generate a random number when chosen.

Continue until the user chooses to exit. Use proper functions and comments.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  Simple Math Utility Program

  ***** Math Utility Menu *****
    1. Square Of A Number
    2. Power Of A Number
    3. Generate A Random Number (1-100)
    4. Exit
  Enter Your Choice (1/2/3/4):3
  Random Number Generated: 26

  ***** Math Utility Menu *****
    1. Square Of A Number
    2. Power Of A Number
    3. Generate A Random Number (1-100)
    4. Exit
  Enter Your Choice (1/2/3/4):4
  Exiting... Goodbye!
○ PS D:\Mountreach_ITR_2026>
```